GOOD DIAGRAMS: EMBEDDING A FINITE TYPE MORPHISM INTO A FINITE LEVEL SQUARE

THE STACKS PROJECT AUTHORS

Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Finite type closed in fp. Limits, Lemma lemma-finite-type-closed-in-finite-presentation (same chapter)
- Descend finite presentation. Limits, Lemma lemma-descend-finite-presentation (same chapter)

Source reference. Original reference 0CNM: sources/stacks-chow/tags.txt, source lines 15897–15897. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZG: sources/stacks-chow/tags.txt, source lines 2591–2591. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 0CNM. Let $X \to S$ be quasi-separated and of finite type. Then there exists an $i \in I$ and a diagram*

$$\begin{array}{ccc} X & \longrightarrow & W \\ \downarrow & & \downarrow \\ S & \longrightarrow & S_i \end{array} \tag{0.1.1}$$

such that $W \to S_i$ is of finite type and such that the induced morphism $X \to S \times_{S_i} W$ is a closed immersion.

Proof. By Lemma 01ZG we can find a closed immersion $X \to X'$ over S where X' is a scheme of finite presentation over S. By Lemma 01ZM we can find an i and a morphism of finite presentation $X'_i \to S_i$ whose pull back is X'. Set $W = X'_i$. □

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